

## Andrew Wei - Senior Software Engineer

San Francisco Bay Area

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### Professional Summary

Military veteran turned full-stack engineering development. Passionate about applying organizational leadership skills to technical domains, especially in product development and fast-paced environments.

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### Professional Experience

#### Human Interest

**Software Engineer** | October 2023 - Present | San Francisco, CA

- Developed full-stack applications using **TypeScript, GraphQL, and React** to enhance financial tools.
  - Optimized system performance by implementing **efficient API structures** and state management.
  - Collaborated cross-functionally to improve **user experience and product features**.
  - Introduced **automated testing strategies**, increasing code coverage and stability.
  - Mentored junior engineers in best practices for **frontend and backend development**.
  - Implemented **secure authentication mechanisms**, ensuring data privacy and compliance.
  - Assisted in architecting **scalable microservices**, improving platform resilience.
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#### Hifive

**Head of Engineering** | October 2022 - September 2023

- Led a team of **3 engineers** in customer-focused product development.
- Designed and implemented **high-performance APIs** with **Ruby on Rails**.
- Integrated **third-party payment providers**, improving financial transaction efficiency.
- Optimized **database queries** for performance and cost savings.
- Spearheaded **security audits** to comply with data protection regulations.
- Implemented **continuous integration and deployment pipelines**.
- Conducted **code reviews and team mentorship**, fostering growth and best practices.

## Software Engineer | June 2021 - October 2022

- Built backend infrastructure and **Google API integrations**.
  - Designed and deployed **AI-assisted message drafting** using **OpenAI APIs**.
  - Developed **async job processing queues** for scalable operations.
  - Created CI/CD pipelines for **automated testing and deployment**.
  - Used **Ruby w/ Roda, Sequel, JavaScript, Svelte, Inertia.js** for full-stack development.
  - Improved application response time by **30%** through backend optimizations.
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## Affirm, Inc.

### Infrastructure Security Engineer | September 2019 - October 2020

- Built **security monitoring ELK stack**, ensuring **real-time threat detection**.
- Managed **PCI compliance** for AWS infrastructure.
- Automated **cloud security monitoring** using Terraform and SaltStack.
- Designed **role-based access control policies** across multiple environments.
- Worked with **Docker/Kubernetes** for containerized application security.
- Conducted **penetration testing and vulnerability assessments**.
- Developed **infrastructure as code (IaC) templates**, improving deployment consistency.

### Analyst, Credit Risk | February 2018 - September 2019

- Conducted **valuation modeling and financial forecasting**.
  - Managed **loan pricing strategies** for Affirm's merchant pricing team.
  - Built **machine learning models** for risk assessment using **scikit-learn**.
  - Optimized database performance on **PostgreSQL and AWS Redshift**.
  - Developed **interactive dashboards** using Looker and Apache Superset.
  - Automated **ETL pipelines** for structured data analysis.
  - Utilized **dbt for data transformation and analytics**.
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## US Army

### Analytics Manager | June 2015 - February 2018

- Led a team of **5 analysts** in network security and data analytics.
- Conducted **threat modeling and security research** using **Wireshark, nmap**.
- Developed **intelligence reports** to support national security operations.
- Used **Python scripting** to automate data analysis workflows.
- Designed **network segmentation strategies** for secure operations.
- Conducted **open-source intelligence gathering (OSINT)**.
- Led **security awareness training** for military personnel.

## Staff Operations Officer | February 2014 - May 2015

- Managed **logistics for training exercises** involving 450+ personnel.
- Coordinated **large-scale troop deployments** across multiple locations.
- Supervised **mission-critical operations** under high-pressure environments.
- Developed **strategic planning reports** for leadership decision-making.
- Ensured **cross-functional team coordination** between departments.
- Led **after-action reviews and performance evaluations**.
- Implemented **efficiency improvements**, reducing operational delays.

## Scout Platoon Leader | September 2011 - January 2014

- Led a **23-member reconnaissance platoon** in combat readiness training.
- Managed **platoon logistics, training, and mission planning**.
- Developed **battlefield simulation strategies** for tactical improvements.
- Ranked **top platoon leader** out of 8 in squadron.
- Ensured **unit preparedness** through rigorous training exercises.
- Collaborated with **senior leadership** for mission execution.
- Awarded **commendations for outstanding leadership and operational excellence**.

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## Education

### United States Military Academy at West Point

*Bachelor of Science (BS) in Computer Science (2007 - 2011)*

### Georgia Institute of Technology

*Master of Science (MS) in Computer Science (2018 - 2021)*

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## Certifications

- **OSCP (Offensive Security Certified Professional)**
- **Network+**
- **Security+**
- **CISSP (Certified Information Systems Security Professional)**
- **Certified Ethical Hacker (CEH)**

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## Technical Skills

- **Programming:** Python, Ruby, JavaScript, TypeScript
- **Frameworks:** Ruby on Rails, React, GraphQL, FastAPI

- **Cloud & DevOps:** AWS, Docker, Kubernetes, Terraform, SaltStack
- **Data Science & Security:** NumPy, Pandas, Scikit-learn, Machine Learning, Threat Modeling
- **Infrastructure & Databases:** PostgreSQL, AWS Redshift, ELK Stack
- **Development Methodologies:** Agile, CI/CD, Secure Coding Practices